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L05: CNN_Chihuahua or Muffin Workshop Reflection
ITA1378 Computer Vision AI
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Lab 05: CNN - Chihuahua or Muffin Workshop Reflective Journal

CNN Architecture

This lab used a convolutional neural network (CNN) to classify images as either a Chihuahua or a muffin. The CNN differed significantly from the traditional fully connected neural network used in Lab 04.

The model contained three convolutional layers. The number of channels increased from 32 to 64 to 128, with each convolution followed by a ReLU activation and max-pooling layer. These pooling operations gradually reduced the spatial dimensions of the images from 224 × 224 to 112 × 112, then 56 × 56, and finally 28 × 28. After the convolutional portion of the network, the learned feature maps were flattened and passed through a fully connected layer with 512 units, followed by dropout and a final output layer containing two classes.

The most important difference from Lab 04 is that the CNN does not immediately flatten the image into one large vector. Instead, the convolutional layers preserve spatial relationships and learn local image features such as edges, textures, and shapes. This makes CNNs better suited for image classification because the arrangement of pixels matters in visual data.

The lab also used data augmentation on the training set through random horizontal flips and rotations. These transformations introduced more variation into the training images and helped reduce the chance that the model would simply memorize the exact appearance of the training examples.

Model Performance

The baseline CNN was trained for 10 epochs using CrossEntropyLoss and the Adam optimizer with a learning rate of 0.001. The model reached a final training accuracy of 96.67% and a final validation accuracy of 96.67%.

The validation set contained 30 images, so this result means that the model correctly classified 29 of the 30 validation images. The only visible misclassification was a Chihuahua predicted as a muffin, while the visible muffin images were classified correctly.

This result was especially interesting because the remaining error still supported the original premise of the dataset: certain Chihuahua faces can visually resemble muffins. Even with a much stronger model, one Chihuahua was still difficult enough to confuse the classifier.

The CNN also showed much stronger generalization than the model used in Lab 04. Validation accuracy reached 96.67% as early as Epoch 5 and again at Epoch 10. Although the accuracy fluctuated somewhat during training, the final training and validation accuracy were equal, suggesting that the model generalized well to unseen images.

Comparison with the Traditional Neural Network

The CNN clearly outperformed the traditional fully connected neural network used in Lab 04.

In Lab 04, the basic neural network reached approximately 90% training accuracy, but validation accuracy peaked around 83.33% and later dropped to approximately 73.33%. This suggested that the model was beginning to overfit the training set.

By comparison, the CNN achieved 96.67% validation accuracy and made far fewer classification errors. The difference illustrates the advantage of using convolutional layers for visual data. The fully connected network in Lab 04 flattened the image immediately, while the CNN retained spatial structure and learned hierarchical visual features before performing classification.

The CNN did require more training time. Each epoch took noticeably longer because convolutional operations and the larger number of model parameters increased computational cost. However, the increase in training time resulted in substantially better validation performance.

Learning Rate Experiment

As part of the required experimentation, I reduced the Adam optimizer learning rate from 0.001 to 0.0001 while keeping the architecture and the number of epochs unchanged.

With the lower learning rate, final training accuracy increased to 99.17%, but validation accuracy decreased to 93.33%. Final validation loss was 0.0985.

This experiment showed that a smaller learning rate did not improve generalization. Although the model fit the training data more closely, performance on the validation set was slightly worse. Because the validation set contained only 30 images, one additional incorrect classification changes the accuracy by approximately 3.33 percentage points. Therefore, the difference should not be overinterpreted, but the original learning rate of 0.001 produced the stronger overall validation result in this experiment.

The experiment reinforced the idea that better training accuracy does not necessarily mean a better model. The most useful performance measure is how well the model performs on unseen data.

Challenges and Solutions

The main challenge in this lab was recognizing which parts of the notebook required modification and which were already complete. Some cells included placeholder values that had to be filled in, such as the input image dimensions.

Another challenge occurred when the notebook could not find the `data` directory. This happened because the GitHub repository had not yet been cloned into the Colab runtime. Cloning the full repository resolved the issue and allowed the notebook to access the training and validation folders.

The notebook also produced warnings related to the number of DataLoader workers. These warnings did not stop the code from running, so I learned to distinguish between warnings and actual errors.

Real-World Applications

CNN-based image classification has many practical uses. Similar systems are used in manufacturing for quality inspection, healthcare for medical image analysis, agriculture for identifying crop disease, retail for product recognition, recycling systems for material sorting, and security applications for visual monitoring.

The structure used in this lab is simplified, but the same general process applies in larger real-world systems: collect labeled images, preprocess the data, train a model, evaluate it on unseen examples, and analyze errors before deployment.

The Chihuahua-or-muffin example also demonstrates why image classifiers must be tested carefully. If visually similar objects can confuse a model in a low-risk example, similar errors in higher-risk systems could have much more serious consequences.

Ethical Considerations

One ethical concern with image classification systems is the risk of deploying models that appear accurate overall but perform poorly on certain categories or types of images. A single accuracy score may hide class-specific weaknesses.

Another concern is overconfidence. In Lab 04, some incorrect predictions were made with very high confidence. This demonstrates why confidence scores should not automatically be interpreted as reliability.

Bias in the training dataset is also important. The dataset in this lab was slightly imbalanced, with more Chihuahua images than muffin images in both the training and validation sets. Although the imbalance was small, real-world datasets can contain much larger representation differences, which may cause models to perform better on some groups than others.

For high-impact applications, developers should evaluate models using multiple metrics, examine misclassifications, monitor performance after deployment, and include appropriate human oversight.

Conclusion

This lab demonstrated why CNNs are more appropriate for image classification than basic fully connected networks. The CNN preserved spatial information, learned visual features through convolutional layers, and achieved substantially better validation performance than the model used in Lab 04.

The learning-rate experiment also showed that changing a hyperparameter does not automatically improve a model. Reducing the learning rate increased training accuracy but slightly reduced validation accuracy, reinforcing the importance of evaluating models based on generalization rather than training performance alone.

The most memorable result was that even the stronger CNN still misclassified one Chihuahua as a muffin. That small error made the lesson surprisingly clear: better architecture can greatly improve performance, but visually ambiguous examples can still challenge a model.

References

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